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Acute Kidney Injury or Physiological Response? Interpretation of Serum Creatinine Elevation After Strenuous Exercise Using Functional and Structural Kidney Biomarkers: A Narrative Review

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ABSTRACT

Background. Serum creatinine commonly rises after strenuous exercise and does not always reflect impaired kidney function. This raises the question of whether post-exercise elevation represents acute kidney injury (AKI) or a transient physiological response.

Aim. To examine post-exercise creatinine elevation and whether additional kidney biomarkers may clarify its significance.

Materials and methods. A narrative review of studies on exercise-related changes in serum creatinine, cystatin C, and kidney injury and stress biomarkers was conducted, considering exercise and hydration.

Results. Post-exercise creatinine elevation may be influenced by muscle mass, muscle metabolism, muscle damage, and hydration and does not always indicate structural kidney injury. Cystatin C is less influenced by muscle mass but reflects filtration rather than structural injury. KIM-1, NGAL, and other biomarkers may add information, although they can also change transiently after exercise and depend on exercise conditions, hydration, and sampling time.

Conclusions. A rise in serum creatinine after strenuous exercise is insufficient evidence of structural kidney injury. Interpretation should consider the time course of changes, hydration, exercise type and intensity, urinalysis, and additional kidney markers when available. No single biomarker can currently make this distinction reliably.

Keywords: acute kidney injury, exercise, creatinine, cystatin C, kidney biomarkers, KIM-1, NGAL

1. Introduction

Acute kidney injury (AKI) is characterized by a rapid decline in kidney function. In practice, the diagnosis relies on serum creatinine and urine output. Under the KDIGO criteria, a rise in serum creatinine of at least 0.3 mg/dL (26.5 μ mol/L) within 48 hours is sufficient to meet the definition of AKI. The same applies when creatinine reaches 1.5 times or more of the baseline value within 7 days. AKI can also be diagnosed based on reduced urine output, specifically when it falls below 0.5 mL/kg/h for at least 6 hours [1,2]. The extent of the increase in creatinine, together with the severity and duration of oliguria, is subsequently used to stage AKI [2]. Creatinine therefore remains one of the primary parameters used in the diagnosis of AKI; however, it primarily reflects changes in kidney filtration function and is not a direct marker of structural kidney injury [1,2]. This is particularly relevant in situations where creatinine levels may be affected by factors other than kidney injury itself, such as strenuous physical exercise.

Physical exercise induces a range of hemodynamic and metabolic changes that also affect kidney function. During exercise, blood flow is redistributed and hemodynamic changes occur, involving both the renal circulation and glomerular filtration [3]. The magnitude of the renal response depends, among other factors, on exercise intensity and duration, as well as environmental conditions and hydration status [3]. Prolonged exercise in high temperatures may be particularly relevant, as fluid loss places additional strain on the mechanisms that maintain adequate renal perfusion. However, the renal response is not uniform across all types of physical activity. In the study by Kawakami et al., high-intensity intermittent exercise did not significantly reduce renal blood flow, suggesting that high exercise intensity alone does not necessarily impair renal perfusion [4].

Hydration status is another important factor that may modify the renal response to exercise. Juett et al. demonstrated that hypohydration during high-intensity intermittent exercise was associated with a greater increase in urinary kidney injury molecule-1 (KIM-1) compared with exercise performed under euhydrated conditions [5]. In a subsequent study by the same group, conducted during prolonged cycling in the heat, hypohydration was also accompanied by greater changes in selected kidney biomarkers [6]. These findings suggest that exercise-induced changes in parameters related to kidney function and injury may depend not only on exercise itself, but also on its intensity and duration, as well as accompanying dehydration and heat stress [3–6].

Serum creatinine can be difficult to interpret in physically active people, particularly in those with greater muscle mass. Since creatinine is derived from creatine metabolism in

skeletal muscle, its serum concentration is partly related to an individual's muscle mass. This may affect estimates of kidney function that rely on creatinine. Groothof et al. showed that the influence of muscle mass was much more evident in creatinine-based estimates of GFR than in estimates based on cystatin C [7]. This issue may be particularly relevant in individuals with above-average muscle mass. Ashouri et al., studying men with testosterone-induced muscle hypertrophy, reported differences in the assessment of kidney function depending on whether creatinine or cystatin C was used [8]. Similarly, Ozkurt et al. highlighted the difficulties associated with assessing kidney function in bodybuilders using anabolic-androgenic steroids and dietary supplements [9]. Although these populations are not representative of all athletes, these studies demonstrate that high muscle mass may complicate the interpretation of creatinine-based kidney function parameters.

Serum creatinine may also be influenced by strenuous exercise and the associated muscle damage. Spada et al. demonstrated that high-intensity resistance training in healthy individuals resulted in increased markers of muscle damage as well as changes in parameters used to assess acute kidney injury [10]. This is relevant when interpreting laboratory results obtained shortly after exercise. An increase in serum creatinine leads to a lower eGFR when calculated using creatinine-based equations; however, this change alone does not indicate how much of the observed decrease reflects an actual decline in kidney function and to what extent it results from factors related to exercise and muscle metabolism [7–10]. Therefore, assessing kidney function in athletes solely based on serum creatinine may be insufficient. This raises a central question for the present review: does an increase in serum creatinine following strenuous exercise truly indicate acute kidney injury, or may it, at least in part, represent a transient physiological response to exercise?

The limitations of creatinine support the inclusion of other biomarkers when assessing exercise-induced changes in the kidneys. These markers may provide different types of information, ranging from changes in filtration function to processes associated with cellular stress and kidney injury. In addition to creatinine and creatinine-based eGFR, cystatin C may provide complementary information on kidney function. Its concentration is less dependent on muscle mass, which may be particularly relevant when assessing physically active individuals [7,8,11]. However, functional markers primarily reflect changes in kidney function and, on their own, cannot determine the nature of potential structural injury.

This assessment may be complemented by biomarkers associated with tubular injury, including neutrophil gelatinase-associated lipocalin (NGAL), kidney injury molecule-1 (KIM-1), and liver-type fatty acid-binding protein (L-FABP) [11,12]. Another group includes

biomarkers of cellular stress, such as tissue inhibitor of metalloproteinases-2 (TIMP-2) and insulin-like growth factor-binding protein 7 (IGFBP7), which have been investigated for the assessment of AKI risk [11,12]. However, the interpretation of changes in these biomarkers following exercise remains unclear. In their systematic review of novel AKI biomarkers, Wołyniec et al. found that physical exercise may be accompanied by transient changes in biomarker concentrations, with the response varying depending on the biomarker and the type of exercise [11]. Combining functional markers with biomarkers of kidney injury and stress may therefore provide a more comprehensive characterization of the renal response to strenuous exercise and help clarify the significance of the observed increase in serum creatinine [11,12].

Despite the growing number of studies investigating the renal response to physical exercise, the interpretation of increased serum creatinine following strenuous exercise remains uncertain. In some athletes, changes in serum creatinine may reach values that meet the diagnostic criteria for AKI; however, creatinine alone cannot determine whether these changes are accompanied by actual structural kidney injury. Mansour et al., studying marathon runners, observed both an increase in serum creatinine meeting AKI criteria in a proportion of participants and changes in biomarkers associated with tubular injury and repair [13]. In contrast, Poussel et al. demonstrated that the assessment of AKI following an ultramarathon may vary depending on the kidney function marker used [14]. These findings indicate that the interpretation of changes in kidney-related parameters following strenuous exercise should not rely solely on serum creatinine. Comparing changes in creatinine with other functional markers and biomarkers of kidney injury and stress may help better characterize the renal response to exercise [13,14].

This narrative review aims to examine whether increases in serum creatinine following strenuous physical exercise reflect true acute kidney injury or a transient physiological response, and to evaluate the role of kidney function markers and biomarkers of kidney injury and stress in differentiating between these conditions.

2. Methods (Literature Search)

The literature for this narrative review was searched using PubMed and Google Scholar. The search covered studies on kidney function after physical exercise, with particular attention to changes in serum creatinine and other markers used in the assessment of acute kidney injury. The main terms used during the search were acute kidney injury, exercise, renal function, serum creatinine, eGFR, cystatin C, KIM-1, NGAL, dehydration, hypohydration, heat stress, muscle

damage, and endurance exercise. Searches related to marathon and ultramarathon running and renal hemodynamics were also performed.

Articles were selected when they provided information relevant to the interpretation of kidney function or kidney-related biomarkers after exercise. The main areas of interest were serum creatinine and cystatin C, biomarkers of kidney injury and stress, hydration and heat exposure, exercise intensity and duration, muscle mass, and exercise-induced muscle damage. Original studies and review articles were used together with clinical guidelines and reference sources needed to describe the definition and interpretation of acute kidney injury.

3. Results

3.1 Renal response to physical exercise

3.1.1 Renal hemodynamics during exercise

Physical exercise is associated with hemodynamic changes that also involve the renal circulation. In response to the increased metabolic demands of working muscles, blood flow is redistributed, potentially affecting renal perfusion and glomerular filtration. However, the magnitude of this response is not constant and depends on several factors, including the type, intensity, and duration of exercise, as well as hydration status and environmental conditions [3].

Changes in the renal circulation during exercise are part of the physiological response to increased physical demands. A reduction in renal blood flow may affect kidney filtration; however, the nature and magnitude of this response depend on the conditions under which exercise is performed. Therefore, transient changes in renal hemodynamics during physical activity should not, by themselves, be interpreted as evidence of structural kidney injury [3].

The relationship between exercise type and renal hemodynamics was investigated by Kawakami et al., who compared high-intensity intermittent exercise with moderate-intensity continuous exercise using ultrasound assessment of renal hemodynamics. The authors found no significant changes in renal blood flow or differences in its response between the two exercise conditions [4]. These findings suggest that high exercise intensity alone does not necessarily cause a significant reduction in renal blood flow and that the renal hemodynamic response may depend on the characteristics of the exercise performed.

Renal hemodynamic changes during exercise should therefore be interpreted as part of a broader physiological response rather than in isolation. Their magnitude may be influenced by exercise duration, hydration status, and environmental conditions [3]. These factors are

particularly relevant when interpreting kidney function parameters measured during or shortly after strenuous exercise.

3.1.2 Effects of exercise intensity and duration

The renal response to exercise depends on its intensity, duration, and type, although the relationship between these factors and changes in kidney function is not straightforward. More pronounced changes in kidney function parameters and biomarkers have been observed during prolonged exercise, whereas high intensity alone is not necessarily associated with significant impairment of renal hemodynamics [3,4,15,16].

The findings of Kawakami et al. illustrate this distinction. Despite differences in exercise intensity between high-intensity intermittent and moderate-intensity continuous exercise, no significant difference in the renal blood flow response was observed [4]. This suggests that exercise intensity should not be considered in isolation when evaluating the renal response to physical activity.

Exercise duration is another important factor to consider. Bongers et al. compared short-duration and prolonged exercise and assessed changes in kidney function and kidney injury biomarkers [15]. In another study by the same group, the response of kidney injury markers to repeated bouts of moderate-intensity endurance exercise was evaluated. After the first day of exercise, uNGAL and uKIM-1 increased, whereas changes in some biomarkers were less pronounced during subsequent exercise sessions [16]. These findings suggest that the renal biomarker response to repeated exercise does not necessarily increase with successive exercise bouts.

Studies conducted during marathons and ultramarathons provide particularly relevant information on the effects of prolonged exercise on kidney function. Several hours of endurance exercise represent a complex physiological challenge and may be accompanied by fluid loss, heat stress, and increased muscle metabolism. Studies of long-distance running have reported post-exercise increases in serum creatinine, with changes in some participants reaching levels consistent with creatinine-based criteria for AKI [13,14,17,18]. However, the interpretation of these findings remains uncertain. Poussel et al. showed that the estimated occurrence of AKI following an ultramarathon may vary depending on the marker used to assess kidney function [14]. Therefore, an increase in serum creatinine following prolonged exercise should not automatically be considered evidence of structural kidney injury.

Available evidence therefore suggests that exercise intensity and duration should be considered together rather than in isolation. Prolonged or very intense exercise may result in

transient changes in parameters used to assess kidney function and injury, although the magnitude and significance of these changes also depend on the type of activity and accompanying conditions [3,4,15,16]. A key issue is distinguishing a transient response to substantial physical exertion from true kidney injury. For this reason, post-exercise changes should be interpreted using serum creatinine alongside other markers of kidney function and injury.

3.1.3 Dehydration and heat stress

Hydration status is an important factor modifying the renal response to physical exercise. Fluid loss during physical activity, particularly when exercise is prolonged or performed in high temperatures, may lead to hypohydration and affect both measures of kidney function and biomarkers of kidney injury. In practice, the effects of hypohydration are difficult to separate from those of exercise intensity and duration, as these factors often occur simultaneously [5,6,15].

The importance of hydration status is well illustrated by the study by Juett et al., in which high-intensity intermittent exercise was performed under hypohydrated and euhydrated conditions. A greater increase in KIM-1 was observed under hypohydrated conditions [5]. This suggests that fluid deficit may increase the response of certain kidney biomarkers to exercise, although the magnitude of the change may vary between biomarkers.

The combination of hypohydration and exercise performed in a hot environment may be particularly relevant. In a subsequent study by Juett et al., participants performed up to 120 minutes of moderate-intensity cycling at an ambient temperature of approximately 35°C. Under hypohydrated conditions, greater changes in osmolality-corrected KIM-1 concentrations and a greater percentage increase in NGAL were observed than under euhydrated conditions [6]. These findings indicate that fluid deficit during exercise in the heat may enhance the response of biomarkers associated with tubular kidney injury.

The relationship between exercise duration, fluid loss, and the renal response was also investigated by Bongers et al. In their comparison of short-duration and prolonged exercise, greater hypohydration during prolonged exercise was accompanied by more pronounced changes in kidney function parameters and kidney injury biomarkers [15]. Because longer exercise duration was accompanied by greater fluid loss, the observed changes cannot be attributed solely to either factor. This highlights the complex nature of the renal response to prolonged physical activity.

The relevance of hypohydration is also apparent under real-world ultraendurance conditions. Linder et al., studying participants in the 100-mile Western States Endurance Run, observed post-exercise increases in kidney biomarkers, with hypohydration associated with greater increases in selected kidney injury biomarkers [19]. These findings indicate that hydration status should also be considered when interpreting changes in kidney-related parameters during prolonged endurance events.

Hypohydration and heat stress should therefore be taken into account when assessing the renal response to intense or prolonged exercise. Available studies indicate that fluid deficit may increase changes in some kidney biomarkers, particularly when combined with prolonged exercise or exposure to high temperatures [5,6,15,19]. At the same time, these changes should not automatically be interpreted as clinically significant AKI. Their interpretation requires consideration of the type and duration of exercise, environmental conditions, and the degree of hypohydration.

3.2 Limitations of Serum Creatinine in Physically Active Individuals

3.2.1 Muscle Mass and Creatinine Production

Serum creatinine is commonly used to assess kidney function, but its concentration is influenced by more than glomerular filtration alone. One important factor is muscle mass, as creatinine is produced through creatine metabolism in skeletal muscle. Therefore, individuals with greater muscle mass may have higher serum creatinine concentrations even in the absence of an actual decline in kidney function [3,7,8].

The importance of muscle mass when interpreting creatinine was demonstrated by Groothof et al. The authors showed that differences in muscle mass may affect the accuracy of creatinine-based estimates of kidney function. This association is considerably weaker for cystatin C, which may be particularly relevant in individuals whose muscle mass differs substantially from average [7]. This issue is especially important in physically active populations, in whom the use of standard creatinine-based parameters may lead to misinterpretation of kidney function.

A similar problem was addressed by Ashouri et al., who assessed kidney function in men with testosterone-induced muscle hypertrophy. The authors compared creatinine- and cystatin C-based parameters, highlighting the potential limitations of creatinine in individuals with increased muscle mass [8]. Although this population is not representative of all athletes, the study illustrates the importance of body composition when interpreting kidney function parameters.

This issue also applies to bodybuilders. Ozkurt et al. assessed kidney function in bodybuilders using anabolic-androgenic steroids and dietary supplements [9]. In this population, the interpretation of kidney-related parameters is particularly challenging because their values may be influenced simultaneously by high muscle mass and other factors related to the substances used and dietary practices.

Muscle mass therefore represents an important limitation when serum creatinine is used to assess kidney function in physically active individuals. Higher creatinine concentrations in this population do not necessarily correspond to a proportional reduction in glomerular filtration, and the results should therefore be interpreted in the context of body composition [3,7–9]. This issue becomes even more complex immediately after strenuous exercise, when serum creatinine may also be affected by acute changes in muscle metabolism and by muscle damage.

3.2.2 Exercise-Induced Muscle Damage

Strenuous physical exercise may lead to transient skeletal muscle damage and alterations in muscle metabolism. This is relevant to the interpretation of serum creatinine because its concentration is not entirely independent of processes occurring within skeletal muscle. Under conditions of substantial exercise stress, a post-exercise increase in creatinine may therefore result from several overlapping mechanisms rather than solely from reduced glomerular filtration [3,10].

This relationship is well illustrated by the study of Spada et al., in which healthy individuals underwent high-intensity resistance training. Exercise was accompanied by evidence of muscle damage as well as changes in parameters used to assess acute kidney injury [10]. This demonstrates that, following strenuous exercise, changes in serum creatinine may occur simultaneously with a marked muscular response, limiting the interpretation of creatinine as an isolated indicator of changes in kidney function.

A similar issue arises during prolonged endurance exercise. Belli et al., studying participants in a mountain ultramarathon, observed concurrent changes related to muscle damage, inflammatory responses, and the risk of acute kidney injury [18]. Under such conditions, changes in serum creatinine occur within a complex systemic response to extreme exercise that involves not only skeletal muscle but also the cardiovascular system and the kidneys.

The complexity of this issue is also highlighted by the systematic review by Wołyniec et al., which examined changes in novel AKI biomarkers following physical exercise. The

responses of individual biomarkers were heterogeneous, emphasizing the difficulty of interpreting post-exercise changes [11].

Muscle damage is therefore one of the factors that complicate the interpretation of a post-exercise increase in serum creatinine as an indicator of altered kidney function. This is particularly relevant after high-intensity resistance training and prolonged endurance exercise [10,18]. Consequently, an increase in serum creatinine immediately following substantial exercise stress should be interpreted within the context of the overall physiological response rather than regarded as independent evidence of structural kidney injury.

3.2.3 Post-Exercise Changes in Serum Creatinine and eGFR

Post-exercise increases in serum creatinine have been observed in various models of strenuous and prolonged exercise, particularly following marathons and ultramarathons [13–15,17,18]. Because creatinine is used in commonly applied equations to estimate glomerular filtration rate, an increase in serum creatinine also leads to a lower creatinine-based eGFR. However, such findings require cautious interpretation when obtained immediately after exercise.

Bongers et al. demonstrated that changes in kidney function and injury parameters may differ depending on the characteristics and duration of exercise and the degree of accompanying dehydration [15]. Similarly, Jouffroy et al., studying participants in an ultra-distance race, observed changes that met AKI criteria during prolonged endurance exercise [17].

Interpretation of the post-exercise decline in eGFR is further complicated by how eGFR is calculated. It represents an estimate based on the concentration of a specific filtration marker rather than a direct measurement of glomerular filtration. Therefore, if serum creatinine changes after exercise because of factors other than filtration itself, creatinine-based eGFR will change accordingly. The effects of muscle mass and muscle metabolism on serum creatinine are particularly relevant in this context [3,7,8].

Poussel et al. highlighted an additional problem: the estimated occurrence of post-exercise AKI may vary depending on the marker used to assess kidney function [14]. Thus, post-exercise changes in serum creatinine may not always correspond to changes observed using markers that are less dependent on muscle metabolism.

Post-exercise increases in serum creatinine and the corresponding reduction in eGFR should therefore be interpreted with consideration of the type and duration of exercise, muscle mass, muscular response to exercise, and hydration status [3,7,8,14,15]. A result obtained immediately after strenuous exercise cannot, by itself, establish whether the observed change

represents true structural kidney injury. This issue becomes particularly important when the increase in serum creatinine reaches the threshold used to define AKI.

3.2.4 Does Creatinine-Defined AKI Represent True Kidney Injury?

Strenuous or prolonged exercise can cause serum creatinine to rise to levels consistent with AKI. Whether this represents actual kidney injury is less certain. However, interpretation of such findings in physically active individuals is not straightforward. Creatinine primarily reflects changes in filtration function and is not a direct marker of structural kidney injury. Furthermore, during exercise, its concentration may be influenced by factors related to muscle metabolism, muscle damage, and hydration status [2,3,11,14]. Therefore, meeting creatinine-based criteria for AKI does not, by itself, demonstrate the presence of structural kidney injury.

This problem is particularly evident in studies of long-distance endurance exercise. Following marathons and ultramarathons, increases in serum creatinine have been reported, with some participants reaching values that fulfil creatinine-based criteria for AKI [13,14,17,18]. Jouffroy et al. observed changes meeting AKI criteria during an ultra-distance race [17]. Belli et al., in turn, reported concurrent changes in parameters related to kidney function, muscle damage, and inflammatory responses during an ultramarathon [18]. These findings demonstrate that changes in serum creatinine during extreme exercise occur within a much broader physiological response.

Particularly important evidence comes from the study by Mansour et al. in marathon runners. The investigators assessed not only serum creatinine but also urine sediment and biomarkers associated with kidney injury and repair. In a substantial proportion of participants, the post-marathon increase in serum creatinine fulfilled AKI criteria, while concurrent changes in biomarkers and urine sediment suggested that, at least in some cases, the response extended beyond an isolated change in serum creatinine [13].

This does not mean, however, that every case of creatinine-defined AKI following exercise represents the same degree of kidney injury. Poussel et al. demonstrated that the assessment of post-exercise kidney function may vary according to the marker used [14]. This is particularly relevant during exercise, when changes in serum creatinine do not necessarily result solely from a reduction in glomerular filtration.

Conversely, a post-exercise increase in serum creatinine should not automatically be regarded as an entirely physiological and clinically irrelevant phenomenon. Studies using biomarkers other than creatinine indicate that strenuous exercise may be accompanied by changes in markers associated with kidney injury or stress [5,6,10,11,13,19]. The systematic

review by Wołyniec et al. confirms that changes in several novel AKI biomarkers have been observed after exercise, although their responses vary according to the biomarker studied and the type of exercise [11].

The most appropriate approach is therefore to regard a post-exercise increase in serum creatinine as a finding that requires interpretation within the appropriate clinical and physiological context rather than as independent evidence of structural kidney injury. Combined assessment of creatinine, functional markers, and biomarkers of kidney injury and stress may provide a more comprehensive picture of the renal response to exercise [3,11–14]. Distinguishing a transient exercise-related response from true kidney injury remains one of the central challenges in the assessment of kidney function in individuals undertaking strenuous physical activity. This issue also provides the basis for discussing alternative biomarkers in the following section.

3.3 Alternative Biomarkers for Assessing Kidney Response to Exercise

3.3.1 Cystatin C

The limitations of creatinine in the assessment of kidney function in physically active individuals support the use of markers that are less dependent on muscle mass. One such marker is cystatin C, which, like creatinine, can be used to estimate GFR. However, its concentration is considerably less dependent on muscle mass, which may be particularly relevant in athletes and other individuals whose body composition differs from that of the general population [7,8,11].

The effect of muscle mass on the assessment of kidney function was demonstrated by Groothof et al. The authors showed that differences in muscle mass may lead to errors in creatinine-based estimates of GFR, whereas estimates based on cystatin C are much less affected by this factor [7]. Although the study was not conducted specifically in athletes, its findings illustrate an important limitation of creatinine-based assessment in individuals whose muscle mass differs substantially from that typically observed in the general population.

This issue was also examined by Ashouri et al. in men with testosterone-induced muscle hypertrophy, including both recreational and competitive athletes [8]. The authors compared cystatin C and creatinine as markers of kidney function and highlighted the potential usefulness of cystatin C when the interpretation of creatinine is complicated by increased muscle mass. However, because of the specific characteristics of the study population, these findings should not be directly generalized to all physically active individuals.

The relevance of cystatin C in the context of physical exercise was also discussed by Wołyniec et al. in their systematic review of changes in AKI biomarkers following exercise [11]. Cystatin C was considered primarily a marker of kidney function, in contrast to biomarkers associated with tubular injury, such as NGAL. Its lower dependence on muscle mass may therefore be particularly useful when assessing kidney function in athletes, in whom interpretation of serum creatinine may be difficult [11]. This does not mean, however, that cystatin C remains unchanged in response to substantial physical stress.

This is illustrated by the study of Bongers et al., who compared the renal response to acute and prolonged exercise [15]. No significant change in cystatin C-based eGFR was observed after acute exercise, whereas a significant reduction was found after prolonged exercise [15]. Importantly, the authors did not demonstrate an association between the degree of hypohydration and cystatin C-based eGFR, indicating that the observed decline cannot be attributed solely to fluid loss. These findings suggest that cystatin C may reduce the problem associated with the influence of muscle mass on kidney function assessment while still reflecting transient changes in filtration function associated with prolonged exercise.

Cystatin C may therefore provide a useful complement to creatinine in the assessment of kidney function in physically active individuals, particularly when the interpretation of creatinine is complicated by atypical muscle mass [7,8,11,15]. However, it remains primarily a marker of filtration function and cannot determine by itself whether post-exercise changes in kidney function are accompanied by structural injury. A more comprehensive assessment therefore requires consideration of biomarkers reflecting other aspects of the renal response, including those associated with tubular injury [11].

3.3.2 KIM-1 and NGAL

KIM-1 and NGAL have been studied as markers of tubular injury following exercise, offering a different perspective from creatinine-based assessment. However, the changes reported after exercise are not consistent across studies. They vary with the type of exercise and hydration status, but can also depend on when the sample is collected and how urinary biomarker concentrations are reported [11].

KIM-1 is of particular interest in studies of the post-exercise renal response. In the systematic review by Wołyniec et al., increases in urinary KIM-1 were reported after different forms of exercise. However, the authors also noted that these increases did not always remain significant after correction for urinary creatinine [11]. This is relevant in exercise studies

because hydration status and changes in urine concentration may affect measured biomarker concentrations and, consequently, influence their interpretation.

A similar issue was demonstrated by Bongers et al. After the first day of moderate-intensity endurance exercise, urinary KIM-1 and NGAL concentrations increased, although their interpretation depended on the method used to account for changes in urine concentration [16]. After normalization, the KIM-1 response varied according to the correction method: the increase was no longer significant when corrected for urinary creatinine or cystatin C, whereas it remained significant after osmolality correction. The increase in NGAL was more consistent across the different normalization approaches [16]. These findings illustrate that the interpretation of urinary biomarkers after exercise depends not only on whether their measured concentrations increase, but also on the method used to account for changes in urine concentration.

Hydration status is another factor that should be considered when interpreting KIM-1 and NGAL. Juett et al. showed that hypohydration during exercise may amplify changes in renal biomarkers [5,6]. In a study involving prolonged cycling in the heat, both osmolality-corrected KIM-1 and the relative change in NGAL were greater under hypohydrated conditions than when euhydration was maintained [6]. These observations suggest that post-exercise changes in KIM-1 and NGAL should not be interpreted without considering fluid balance and the environmental conditions in which exercise was performed.

Relevant evidence also comes from studies of prolonged endurance exercise. Mansour et al. reported post-marathon increases in a range of biomarkers associated with kidney injury and repair, including KIM-1 and NGAL [13]. Creatinine, creatine kinase, albuminuria, and urine sediment were assessed at the same time, allowing the biomarker response to be considered alongside changes in conventional measures of kidney function and other exercise-related parameters. These findings demonstrate that post-marathon changes in creatinine may coexist with changes in biomarkers reflecting other aspects of the renal response to exercise.

Changes in biomarkers associated with tubular injury have also been observed after exercise of a different nature. Spada et al. found that a single session of high-intensity resistance exercise was accompanied by increases in markers of muscle damage, a small rise in creatinine, and transient changes in several renal biomarkers, including NGAL [10]. Urinary biomarker changes were most apparent shortly after exercise and returned to baseline values within 24 hours, emphasizing the importance of sampling time when interpreting measurements obtained after strenuous physical activity.

Taken together, the available data suggest that KIM-1 and NGAL may complement creatinine-based assessment because they provide information about aspects of the renal response that are not captured by filtration markers alone. Their increases should not, however, be interpreted independently of the conditions in which they were measured. Several factors can affect KIM-1 and NGAL measurements after exercise, including the type and duration of exercise, hydration status, sampling time, and the way urinary concentrations are normalized [5,6,11,13,16]. For this reason, a rise in either marker should not be viewed as evidence of clinically significant AKI on its own. KIM-1 and NGAL are more informative when considered alongside kidney function markers and in the context of the exercise performed and the individual's clinical condition.

3.3.3 Other Kidney Injury and Stress Biomarkers

KIM-1 and NGAL are not the only biomarkers that have been examined after exercise. Studies have also looked at liver-type fatty acid-binding protein (L-FABP), interleukin-18 (IL-18), tissue inhibitor of metalloproteinases-2 (TIMP-2), and insulin-like growth factor-binding protein 7 (IGFBP7) [11]. These markers reflect different biological processes and may therefore provide information that complements conventional measures of filtration function and more frequently investigated biomarkers of tubular injury.

Evidence regarding L-FABP after exercise remains limited. In the systematic review by Wołynec et al., only two studies in healthy individuals were identified. One reported an increase in urinary L-FABP after incremental maximal cycling exercise, whereas another found no significant change after calf-raise exercise [11]. These limited and inconsistent findings make it difficult to determine the significance of post-exercise changes in L-FABP. Therefore, L-FABP should not be interpreted in isolation as evidence of acute kidney injury.

IL-18 has also been evaluated in exercise studies, with increases reported following physical exercise [11]. Such a response was observed, among other settings, after high-intensity resistance exercise [10]. However, IL-18 is also involved in inflammatory processes, which limits its specificity as an isolated marker of kidney injury. This is particularly relevant following strenuous or prolonged exercise, which may itself induce a systemic inflammatory response that is not confined to the kidneys [10,11]. The interpretation of post-exercise changes in IL-18 therefore requires consideration of the broader physiological response to exercise.

A broader picture of the changes occurring after prolonged exercise was provided by Mansour et al. In marathon runners, the authors assessed biomarkers associated with injury, including IL-6, IL-8, IL-18, KIM-1, NGAL, and TNF- α , as well as repair-associated markers

such as YKL-40 and MCP-1 [13]. Concentrations of these markers increased immediately after the marathon, and most declined over the following 24 hours [13]. Their simultaneous assessment alongside creatinine, albuminuria, and urine sediment demonstrated that the renal response to prolonged exercise may involve both changes in conventional functional parameters and alterations in biomarkers reflecting injury, inflammation, and repair processes.

TIMP-2 and IGFBP7 represent a distinct group of biomarkers associated with cellular stress [11]. They provide information different from that obtained from creatinine, which primarily reflects filtration function, and from biomarkers associated with tubular injury, such as KIM-1 and NGAL. However, evidence regarding the behaviour of TIMP-2 and IGFBP7 immediately after exercise remains limited. In the systematic review by Wołyniec et al., only one study evaluating these biomarkers in relation to physical exercise was identified [11]. The available evidence is therefore insufficient to determine the significance of their post-exercise changes or to interpret an isolated increase as evidence of clinically significant AKI.

Overall, the available evidence indicates that the renal response to exercise is not limited to changes in creatinine or to increases in individual biomarkers associated with tubular injury. Changes may also occur in markers related to cellular stress, inflammation, and repair processes [11,13]. The significance of these responses remains incompletely understood, particularly because many of the observed changes appear to be transient and because the available studies differ considerably in their methodology. Accordingly, none of these biomarkers should be interpreted in isolation as evidence of clinically significant post-exercise AKI. Their potential value lies rather in providing complementary information about different components of the renal response [11,13].

4. Discussion

This review examined a practical question: how should an increase in serum creatinine after strenuous exercise be interpreted, and can other kidney biomarkers help distinguish a transient physiological response from true kidney injury? The available evidence suggests that this distinction remains difficult. Creatinine is an important marker of kidney function, but after exercise its interpretation may be influenced by muscle mass, muscle breakdown, hydration status, and the intensity and duration of exercise [3,7,8,10,15]. At the same time, findings from studies using urine microscopy and biomarkers of kidney injury suggest that exercise-related changes cannot be assumed to be entirely physiological in every individual [3,10,11,13,15].

4.1 Integrating Functional, Injury, and Stress Biomarkers

The evidence discussed above illustrates why assessment of the renal response to exercise on the basis of a single biomarker is problematic. Functional markers and biomarkers associated with injury or cellular stress reflect different aspects of renal physiology and may respond differently to the same exercise exposure. Creatinine and creatinine-based eGFR primarily provide information about filtration function, but their interpretation in physically active individuals is complicated by muscle mass, muscle metabolism, and exercise itself [3,7,8]. Cystatin C may reduce some of these limitations because it is less dependent on muscle mass, although it also remains primarily a marker of filtration function [7,8]. Neither marker can therefore determine by itself whether a post-exercise change in kidney function is accompanied by structural kidney injury.

Biomarkers such as KIM-1 and NGAL provide complementary information because they are associated with processes other than filtration alone. Exercise studies have demonstrated post-exercise changes in their concentrations, but these responses do not always parallel changes in creatinine or cystatin C and may be influenced by exercise duration, hydration status, sampling time, and the method used to express urinary biomarker concentrations [5,6,10,11,15]. Other biomarkers associated with cellular stress, inflammation, and repair may further broaden this assessment, although the evidence supporting their interpretation in the exercise setting remains more limited [11,13]. These differences underline why functional and injury-related biomarkers should be regarded as complementary rather than interchangeable measures.

Studies combining several types of measurements provide useful examples of this approach. Mansour et al. demonstrated simultaneous post-marathon changes in serum creatinine, urine sediment, and several biomarkers associated with kidney injury and repair [13]. Bongers et al. showed that prolonged exercise was accompanied by changes in both kidney function and injury biomarkers, although individual parameters did not respond in the same manner [15]. Similarly, Linder et al. reported increases in several renal biomarkers following a 100-mile ultramarathon, with the degree of hypohydration associated with the magnitude of some of these changes [19]. Together, these findings demonstrate that the interpretation of the renal response depends not only on which biomarkers are measured but also on the conditions under which exercise is performed.

A more complete assessment of the renal response to strenuous exercise may therefore require integration of several sources of information rather than reliance on a single laboratory parameter. Functional markers such as creatinine and cystatin C may be considered together

with biomarkers associated with tubular injury, such as KIM-1 and NGAL, and, where supported by available evidence, markers of cellular stress and repair [3,11,13]. The interpretation of these measurements should also take into account exercise type and duration, hydration status, and the time course of biomarker changes during the post-exercise period [3,11,15,19]. Such an approach may provide a more complete picture of whether an observed change predominantly reflects a transient alteration in filtration function or is accompanied by biomarker changes suggestive of kidney injury.

At present, however, the available evidence does not support a single biomarker or standardized biomarker panel that can reliably distinguish a physiological response to exercise from clinically significant AKI in all settings. Standardized approaches to the interpretation of many novel biomarkers in exercise are still lacking, and the clinical significance of transient post-exercise changes remains incompletely defined [11]. Biomarker results should therefore be interpreted collectively and in the context of the exercise exposure rather than as isolated evidence for or against acute kidney injury.

4.2 Clinical interpretation and practical implications

4.2.1 Interpreting post-exercise changes in kidney function

The assessment of kidney-related parameters after physical exercise requires consideration of the conditions under which these changes occur. An increase in serum creatinine after intense or prolonged exercise may meet laboratory criteria for acute kidney injury (AKI); however, this does not automatically indicate structural kidney injury. In the post-exercise period, creatinine concentration may be influenced by both changes in kidney filtration and non-renal factors associated with physical activity. Therefore, a single measurement obtained immediately after exercise should be interpreted cautiously and within the appropriate context [3,11].

One of the main limitations of creatinine is its dependence on muscle metabolism. Intense exercise, particularly when accompanied by muscle damage, may increase serum creatinine independently of changes in glomerular filtration. At the same time, prolonged physical activity is associated with changes in renal hemodynamics and may lead to fluid loss. The post-exercise rise in creatinine may therefore reflect several overlapping processes, limiting its usefulness as an isolated marker of acute kidney injury [3,11,15].

This has direct implications for the interpretation of estimated glomerular filtration rate (eGFR). Because creatinine-based equations use serum creatinine to estimate glomerular filtration, a post-exercise increase in creatinine results in a lower calculated eGFR. However,

this change may not fully correspond to an actual reduction in glomerular filtration. The choice of biomarker used to assess kidney function may therefore influence both the estimated magnitude of post-exercise changes in filtration and the frequency with which AKI is identified [11,14].

This issue is well illustrated by the study by Poussel et al. in ultramarathon participants. The authors highlighted the difficulties associated with identifying exercise-induced AKI on the basis of creatinine and demonstrated differences in the assessment of kidney function depending on the biomarker used [14]. These findings indicate that conventional creatinine-based criteria require cautious interpretation in the setting of intense and prolonged exercise.

Nevertheless, an increase in creatinine after exercise should not automatically be regarded as a physiological and clinically insignificant response. Mansour et al. demonstrated that, in addition to an increase in creatinine meeting AKI criteria in a substantial proportion of marathon runners, changes in kidney injury and repair biomarkers and abnormalities in urine sediment consistent with tubular injury were also observed [13]. These findings suggest that, at least in some individuals, the renal response to substantial exercise stress may extend beyond an isolated change in serum creatinine.

A post-exercise increase in creatinine should therefore be regarded as a finding requiring interpretation within a broader context rather than as independent evidence of structural kidney injury. Assessment should consider the time course of abnormalities, the characteristics of exercise and hydration status and, when appropriate, other markers of kidney function and biomarkers of kidney injury [3,11,13–15].

4.2.2 Distinguishing physiological response from true kidney injury

A rise in serum creatinine after exercise can be difficult to interpret. Changes in kidney filtration are one possible explanation, but they are not the only one. Creatinine levels can also be affected by muscle metabolism and muscle damage during exercise, as well as by changes in hydration. Therefore, meeting creatinine-based AKI criteria immediately after intense exercise does not, by itself, establish the presence of structural kidney injury [2,3,11,14,15].

One factor that may assist interpretation is the time course of the observed changes. Hernando et al. followed 76 marathon participants from the day before the race until 192 hours after its completion. Serum creatinine increased immediately after the marathon and subsequently decreased during recovery [20]. This demonstrates that a measurement obtained immediately after exercise may not reflect the subsequent course of kidney function.

However, a rapid decline in serum creatinine does not exclude the possibility of transient kidney injury. In the study by Mansour et al., 82% of marathon runners met creatinine-based AKI criteria after the race, while urine microscopy demonstrated features of tubular injury in 73% of participants. Increases in biomarkers associated with kidney injury and repair were also observed after the marathon [13]. Thus, in some athletes, an increase in creatinine may coexist with other indicators of kidney injury and should not automatically be considered an exclusively physiological phenomenon.

Conversely, an increase in novel biomarkers does not provide an unequivocal answer either. The systematic review by Wołyniec et al. showed that exercise is associated with changes in several biomarkers used in AKI research, but interpretation was complicated by differences in study populations, exercise protocols, and methodological approaches [11]. Some biomarkers may also be influenced by processes that are not specific to the kidneys, including inflammation and hypoxia. Their post-exercise elevation therefore cannot automatically be equated with clinically significant AKI.

The context in which exercise is performed is also important. Exercise duration and intensity, as well as hydration status, appear particularly relevant. Experimental and field studies indicate that hypohydration may augment the increase in selected kidney-related biomarkers following exercise [5,6,19]. In the study by Linder et al., several urinary biomarkers increased following a 100-mile ultramarathon, and the degree of hypohydration was associated with the magnitude of some of these changes [19].

In practice, distinguishing a transient post-exercise response from true kidney injury should therefore not rely on a single parameter. A more appropriate approach is to consider the time course of creatinine and other functional markers together with injury biomarkers, urinalysis findings, and the conditions under which the exercise was performed [3,11,13,19,20]. Such an approach better reflects the complexity of the renal response to exercise than identifying kidney injury solely on the basis of changes in serum creatinine.

4.2.3 Factors modifying the interpretation of post-exercise biomarkers

The interpretation of post-exercise changes in kidney biomarkers requires consideration of the conditions under which these changes occur. The renal response is not uniform across different forms of physical activity, and the magnitude of observed changes may depend on exercise intensity and duration, hydration status, environmental conditions, the extent of muscle damage, and the timing of sample collection [3,11,15].

Exercise intensity and duration are important determinants of the observed response. Bongers et al. demonstrated differences in kidney function parameters and injury biomarkers between shorter exercise and prolonged exercise accompanied by a greater degree of hypohydration [15]. Kawakami et al., in turn, compared high-intensity intermittent exercise with moderate-intensity continuous exercise and assessed renal hemodynamics, kidney function, and biomarkers at several time points [4]. These studies indicate that the characteristics of the exercise stimulus should be considered when interpreting post-exercise results.

Hydration status is another important factor. Juett et al. demonstrated that hypohydration during high-intensity intermittent exercise may augment increases in selected kidney-related biomarkers [5]. Similar associations have been observed during prolonged exercise performed in the heat [6] and following ultramarathon running [19]. Hydration status should therefore be regarded as an important component of the interpretation of post-exercise biomarker changes.

Environmental conditions may also influence the renal response. Exercise in hot conditions can promote fluid loss, and the combination of heat stress and hypohydration may enhance changes in selected kidney biomarkers. In the study by Juett et al., during cycling at approximately 35°C, hypohydration was associated with higher osmolality-corrected urinary KIM-1 concentrations and a greater percentage increase from baseline in osmolality-corrected urinary NGAL compared with maintenance of euhydration [6].

Muscle damage is particularly relevant when interpreting serum creatinine. Following high-intensity resistance exercise, Spada et al. observed increases in creatine kinase and myoglobin together with changes in creatinine and urinary kidney injury biomarkers [10]. This demonstrates that after intense exercise, the muscular response may coexist with changes in parameters used to assess kidney function, further complicating the interpretation of creatinine.

The timing of sample collection is also important because individual biomarkers show different post-exercise kinetics [11]. In the study by Kawakami et al., urinary NGAL corrected for creatinine increased 60 minutes after exercise, although none of the values exceeded the cut-off used by the authors for kidney injury [4]. Thus, both the timing of measurement and the characteristics of the biomarker itself should be considered when interpreting the result.

The kidney response seen after exercise is influenced by more than the exercise itself. Duration and intensity are important, but so are heat exposure and hydration. At the same time, exercise-related muscle damage can raise creatinine and make the result difficult to interpret. Timing is another consideration, as biomarkers measured soon after exercise may show a

different pattern from those measured later during recovery. Post-exercise changes in kidney-related parameters should therefore not be interpreted independently of the circumstances in which the exercise was performed [3,10,11,15,19].

4.2.4 Practical approach to kidney assessment after exercise

Assessment of kidney function after intense exercise should not be based on a single parameter. An increase in serum creatinine should be interpreted in relation to the type and duration of exercise, hydration status, possible muscle damage, and the time course of abnormalities during recovery [3,11,14,15].

Cystatin C may provide useful complementary information. It is less dependent on muscle mass than creatinine, and available evidence suggests that it may provide a more reliable reflection of glomerular filtration than creatinine in physically active individuals under certain conditions [7,8,11]. Nevertheless, cystatin C remains primarily a marker of filtration and cannot by itself determine whether a change in kidney function is accompanied by structural injury.

Biomarkers associated with kidney injury, including KIM-1 and NGAL, provide a different type of information. They may reveal changes that cannot be assessed using creatinine alone, although their interpretation also has important limitations. Post-exercise increases in these biomarkers have been observed in different exercise settings, and their magnitude may depend on factors including hydration status, the timing of sample collection, and the method used to normalize urinary concentrations [5,6,11,15].

The time course of abnormalities is therefore particularly important. In the study by Mansour et al., serum creatinine and the investigated kidney injury and repair biomarkers were highest immediately after the marathon and subsequently decreased during follow-up [13]. Other studies similarly indicate that at least some abnormalities observed immediately after exercise are transient [10,11,20].

Practical interpretation should therefore combine information on filtration with markers of potential kidney injury and the context in which the results were obtained. Serum creatinine and eGFR, cystatin C when appropriate, urinalysis and available injury biomarkers should be considered together with hydration status, exercise characteristics, and the time course of abnormalities [3,11,13,15]. This should not be regarded as a separate diagnostic algorithm, but rather as a framework for interpreting individual parameters as components of a broader clinical picture.

At present, no single biomarker can unequivocally distinguish a transient response to intense exercise from true kidney injury. The most appropriate interpretation therefore involves integrating markers of kidney function with biomarkers of potential injury, their temporal pattern, and the clinical and exercise context [3,11].

4.3 Limitations and Future Perspectives

4.3.1 Methodological Limitations of Current Studies

Current studies investigating the renal response to physical exercise are characterized by considerable methodological heterogeneity, which makes direct comparison of their findings and the formulation of definitive conclusions difficult. Differences concern both the populations studied and the type, intensity, and duration of exercise. Available studies have assessed renal responses following, among others, intermittent exercise, endurance and resistance training, marathons, and ultramarathons [4–6,10,13–19]. Such diversity of exercise protocols means that the observed changes in biomarkers may reflect different degrees and types of physiological stress, and findings obtained in one exercise setting cannot be directly extrapolated to other forms of physical activity.

The conditions under which exercise is performed represent an additional source of variability. Hydration status and exposure to heat may influence the renal response and the concentrations of measured biomarkers [5,6,15,19]. The relevance of these factors has been demonstrated both in controlled experimental settings and during real-world sporting events. In the study by Linder et al., hypohydration augmented the post-exercise increase in some urinary biomarkers of kidney injury following a 100-mile ultramarathon [19].

Another important limitation is the lack of full standardization in biomarker assessment. Studies differ in the biomarkers selected and in the timing of blood and urine sampling, which is particularly relevant because post-exercise changes may be dynamic and transient [11]. Wołyniec et al. emphasized that differences in study populations, designs, and methodologies complicate comparisons between studies, while short follow-up periods limit the ability to determine the clinical significance of abnormalities detected immediately after exercise [11].

The interpretation of urinary biomarkers presents an additional challenge. Exercise and associated changes in fluid balance may alter urine concentration; therefore, raw biomarker concentrations do not necessarily reflect changes occurring exclusively within the kidneys. Different normalization approaches have been used, including adjustment for creatinine, osmolality, urine flow rate, or cystatin C, but each of these methods has limitations [11]. This issue is particularly relevant in exercise studies because creatinine is the most commonly used

reference for normalization, while its urinary excretion may itself change in response to exercise [11].

Methodological heterogeneity does not invalidate the observed biomarker changes, but it limits direct comparison between studies and makes it difficult to establish which patterns of response truly indicate kidney injury. This remains one of the major limitations of the current literature and should be considered when interpreting the findings of individual studies [11].

4.3.2 Uncertainty in Defining Exercise-Induced AKI

One of the fundamental challenges in research on the effects of intense physical exercise on the kidneys is the definition of exercise-induced AKI itself. Conventional diagnostic criteria rely largely on changes in serum creatinine, yet interpretation of this parameter is particularly difficult after intense exercise. An increase in creatinine may reflect a reduction in glomerular filtration, but changes related to muscle metabolism and muscle damage, as well as hydration status, may also contribute. Therefore, meeting creatinine-based criteria for AKI after intense exercise does not necessarily establish the presence of structural kidney injury [2,3,11].

However, the opposite assumption—that every post-exercise increase in creatinine represents a harmless physiological response—should also be avoided. In some studies, changes in functional parameters have been accompanied by findings suggestive of structural kidney injury. Mansour et al. demonstrated that the post-marathon increase in creatinine was accompanied by increases in kidney injury and repair biomarkers, as well as urinary sediment findings indicative of tubular injury. Most of these abnormalities decreased markedly within 24 hours [13]. These findings suggest that, at least in some individuals, the renal response to intense exercise may extend beyond an isolated increase in serum creatinine.

The inclusion of kidney injury biomarkers does not, however, fully resolve the diagnostic uncertainty. KIM-1, NGAL, and other biomarkers may increase after intense or prolonged exercise, but the significance of transient elevations observed in otherwise healthy individuals remains unclear [11,13,14]. Poussel et al. observed an increase in NGAL following an ultramarathon despite the absence of a significant deterioration in GFR [14]. This illustrates why an increase in an injury biomarker alone is insufficient to determine the nature of the post-exercise renal response.

Another important limitation is the lack of standardized cut-off values for novel biomarkers in the context of physical exercise. Cystatin C and biomarkers such as NGAL and KIM-1 may complement the information provided by creatinine, but their role in diagnosing exercise-induced AKI has not yet been clearly established [3,11]. Alternative biomarkers

continue to be used primarily in research settings, partly because standardized cut-off values are lacking and their application in routine clinical practice remains limited [3].

The uncertainty surrounding the definition of exercise-induced AKI therefore reflects limitations of both conventional and novel biomarkers. An isolated increase in creatinine cannot reliably determine whether structural kidney injury has occurred, while a transient increase in injury biomarkers should likewise not be interpreted independently of other findings and their evolution over time [3,11,13,14]. Current evidence does not allow identification of a single biomarker or threshold that can reliably distinguish a physiological post-exercise response from true AKI [3,11].

4.3.3 Directions for Future Research

Future studies should primarily aim to address the methodological issues that currently limit comparisons between investigations. More standardized exercise protocols, assessment of hydration status, and sampling schedules could help characterize the typical time course of renal changes following exercise and facilitate identification of responses that extend beyond transient alterations associated with physiological stress [11]. Larger and more diverse study populations are also needed, as much of the available evidence is derived from relatively small cohorts and widely differing exercise models [11].

An important direction for future research is the simultaneous assessment of markers reflecting different aspects of the renal response. Creatinine primarily provides information about functional changes, and its interpretation immediately after exercise has important limitations. Combining creatinine and cystatin C with biomarkers of kidney injury and stress may provide a more comprehensive characterization of the renal response to exercise [3,11,13]. A multimarker approach was used, for example, by McKenna et al., who assessed creatinine together with NGAL, KIM-1, TIMP-2, and IGFBP7 following a marathon [21]. This does not, however, mean that such a combination currently represents a validated diagnostic panel capable of reliably identifying exercise-induced AKI.

Assessment of the time course of biomarker changes is equally important. A single measurement obtained immediately after exercise cannot determine whether an abnormality is transient or persists during recovery. Future studies should therefore include serial sampling before exercise, immediately after its completion, and at subsequent time points. Such an approach could better characterize the timing of biomarker increases and their return toward baseline, as well as clarify the significance of transient post-exercise changes [11].

Future protocols should also carefully account for hydration status and environmental conditions. Linder et al. demonstrated increases in urinary kidney injury biomarkers following a 100-mile ultramarathon, with hypohydration augmenting the increase in some of these markers [19]. However, these relationships may not be consistent across different exercise settings. In the study by McKenna et al., biomarkers of organ stress and injury increased after the Boston Marathon, but their responses did not differ significantly according to post-race hydration status classification [21]. These findings support the need to evaluate hydration in conjunction with the type and duration of exercise and the environmental conditions in which it is performed.

One of the most important unresolved questions concerns the long-term significance of repeated episodes of post-exercise renal changes. Hoffman and Weiss found that although runners who met creatinine-based AKI criteria after one 161-km ultramarathon frequently met these criteria again after a subsequent race, previous AKI was not associated with greater renal dysfunction following a later exercise stimulus of similar magnitude [22]. However, these findings were based on a relatively small number of runners with repeated measurements and do not establish the long-term consequences of recurrent exercise-associated renal changes. Bongers et al. did not observe a cumulative increase in the investigated kidney injury markers over three consecutive days of moderate-intensity endurance exercise, although the responses of uNGAL and uKIM-1 and their dependence on the method of normalization were not identical [16]. Both studies therefore provide information on the renal response to repeated exercise exposure but cannot determine its long-term consequences.

Long-term prospective studies incorporating serial assessment of kidney function together with biomarkers of injury and stress are therefore needed. Such studies may help determine not only which changes represent a transient response to exercise, but also whether repeated episodes have clinical significance over time. Current evidence does not yet provide a definitive answer to this question [3,11,22], and the long-term significance of recurrent exercise-associated renal changes remains uncertain.

5. Conclusions

An increase in serum creatinine following intense physical exercise should not automatically be equated with structural kidney injury. The interpretation of creatinine in the post-exercise period is complicated by factors that are partly independent of glomerular filtration itself, including muscle metabolism and muscle damage as well as changes in hydration status [3,11,15]. Therefore, a transient increase in creatinine and the corresponding

decrease in creatinine-based eGFR should be interpreted in the context of the type and intensity of exercise and other factors that may influence creatinine concentration. Meeting creatinine-based criteria for AKI is not, by itself, sufficient to confirm structural kidney injury after exercise [3,11]. This interpretation is also consistent with recent literature emphasizing the limitations of creatinine-based definitions of AKI after strenuous exercise [3,11].

Cystatin C may provide a valuable complement to creatinine in the assessment of kidney function in physically active individuals, particularly when unusual muscle mass complicates the interpretation of creatinine. Its concentration is less dependent on muscle mass than serum creatinine, which may reduce an important source of error when assessing kidney function in this population [7,8]. Available evidence on the response to exercise supports the usefulness of cystatin C as a complementary marker of glomerular filtration [11]. However, cystatin C does not fully resolve the problem of interpreting post-exercise changes because it remains primarily a marker of filtration and, on its own, cannot determine whether functional changes are accompanied by structural kidney injury [11,15]. The review by Wołyniec et al. also found more consistent evidence for cystatin C than for the other biomarkers examined [11].

KIM-1 and NGAL can add information that is not provided by creatinine or cystatin C alone. In studies involving intense or prolonged exercise, both markers of kidney function and biomarkers linked to tubular injury have been found to increase after exercise [11,13,15,18,19]. However, these changes do not necessarily follow the same pattern or have the same meaning. KIM-1, NGAL, and other biomarkers can rise for a short period and then return toward baseline, which makes their clinical relevance difficult to determine in otherwise healthy individuals [11,13,14]. A single elevated value after exercise is therefore difficult to interpret on its own and cannot confirm clinically significant AKI [11,13,14].

No single marker provides a complete picture of how the kidneys respond to exercise. Creatinine and cystatin C mainly reflect filtration, whereas markers of kidney injury or stress describe other aspects of the renal response. Looking at these measures together may therefore be more informative than relying on any one of them alone [3,11,13]. Urinalysis and the time course of abnormalities on serial measurements should also be considered, as some changes observed immediately after exercise are transient [11,13]. Interpretation should additionally take into account hydration status and the type, intensity, and duration of exercise, as these factors may influence both the renal response and measured biomarker concentrations [3,15,18,19]. None of these parameters should therefore be interpreted in isolation from the other findings and the overall clinical context. Such an approach may reduce both the risk of

overinterpreting an isolated rise in creatinine as evidence of AKI and the risk of prematurely dismissing all post-exercise abnormalities as a harmless physiological response [3,11,13].

Despite the growing body of evidence, no single biomarker currently allows a reliable distinction between a transient renal response to exercise and true kidney injury [3,11]. Interpretation of the available evidence is further complicated by differences in exercise protocols, study populations, sampling times, and methods used to measure and report biomarkers [11]. The long-term significance of repeated episodes of post-exercise elevations in creatinine and kidney injury biomarkers also remains uncertain, as evidence regarding their potential long-term consequences is limited [3,16,22]. Future studies should therefore use more standardized protocols, include serial measurements of markers of kidney function, injury, and stress, and provide longer follow-up of athletes [3,11]. Such an approach may help determine which changes reflect a transient response to exercise and which may indicate clinically relevant kidney injury [3,11].

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Conceptualization: AKŁ, DG, ŁI;

Methodology: AKŁ, JD, AK;

Software: KL, DG, JG;

Check: JG, JD, DG, KL, AK, ŁI;

Formal analysis: AKŁ, DG, JD;

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Resources: JG, KL, AK, ŁI;

Data curation: AKŁ, JG, KL;

Writing-rough preparation: AKŁ, DG, ŁI;

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